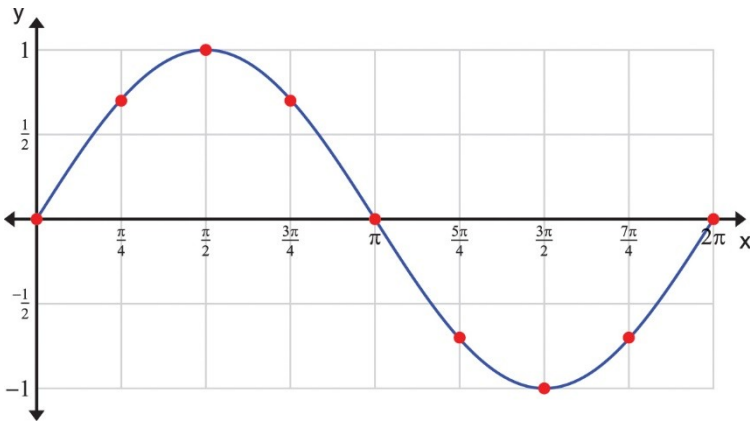
Graph $y = \sin x$ and $y = \cos x$

The table gives several values of x and the corresponding values of $\sin x$.

x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\sin x$	0	$\frac{\sqrt{2}}{2}$	1	$\frac{\sqrt{2}}{2}$	0	$-\frac{\sqrt{2}}{2}$	-1	$-\frac{\sqrt{2}}{2}$	0

Plotting these as ordered pairs gives the graph of $y = \sin x$.



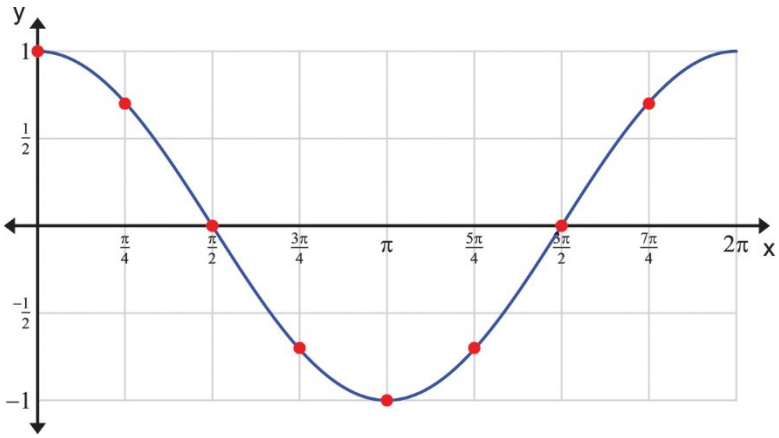
The sine function is periodic with period 2π.

This means that the pattern shown between 0 and 2π repeats infinitely far to the left and right.

The next table gives several values of x and the corresponding values of $\cos x$.

x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\cos x$	1	$\frac{\sqrt{2}}{2}$	0	$-\frac{\sqrt{2}}{2}$	-1	$-\frac{\sqrt{2}}{2}$	0	$\frac{\sqrt{2}}{2}$	1

Plotting these as ordered pairs gives the graph of $y = \cos x$.

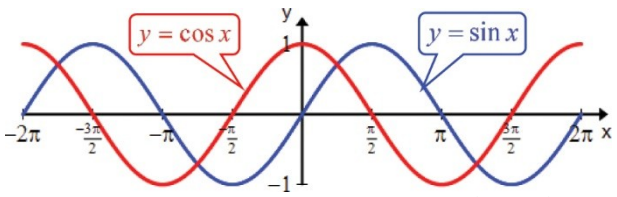


The cosine function is also periodic with period 2π.

The pattern shown between 0 and 2π repeats infinitely far to the left and right.

The graphs of $y = \sin x$ and $y = \cos x$ are shown on the interval $[-2\pi, 2\pi]$ for comparison.

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Characteristics of the graphs of sin *x* and cos *x:* The domain is $(-\infty, \infty)$. The range is $[-1, 1]$. The period is 2π.

The graph of sin *x* is symmetric with respect to the origin.
$y = \sin x$ is an odd function.

The graph of cos *x* is symmetric with respect to the *y*-axis.
$y = \cos x$ is an even function.

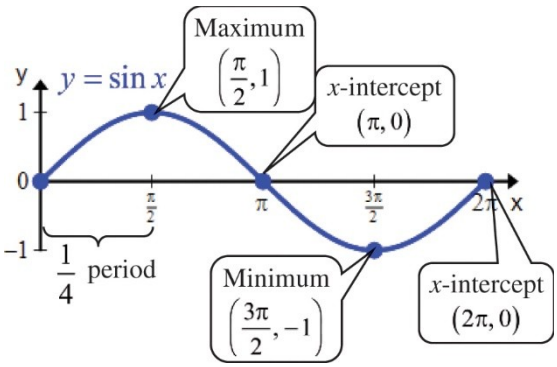
The graphs of sin *x* and cos *x* differ by a horizontal shift of $\frac{\pi}{2}$.

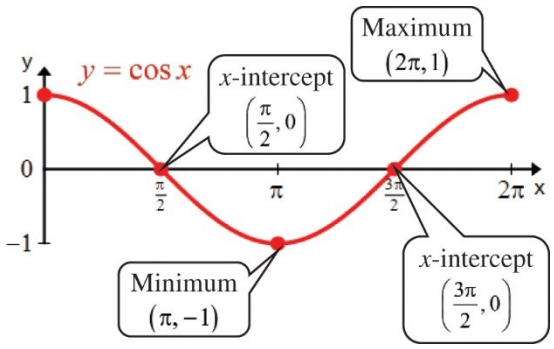
$y = A\sin x$ $y = A\cos x$

Graph and

To graph variations of these functions we may graph the key points - relative minima, relative maxima, and *x*-intercepts.

The graphs of both the sine and cosine function alternate between a relative minimum or maximum and an *x*-intercept for each quarter of a period.





The graph of $y = A \cdot f(x)$ is the graph of $y = f(x)$ with

a vertical stretch if $|A| > 1$
a vertical shrink if $0 < |A| < 1$
a reflection across the *x*-axis if $A < 0$

Amplitude: For the graphs of $y = |A|\sin x$ and $y = |A|\cos x$, the vertical scaling factor $|A|$ is the amplitude.
The amplitude represents the amount of deviation from the central position of the sine wave.

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The amplitude is half the distance between the maximum value of the function and the minimum value of the function.

Identify the amplitude of each function.

1. $f(x) = -8\sin x$

$\text{amp} = |-8| = 8$

2. $g(x) = \frac{2}{3}\cos x$

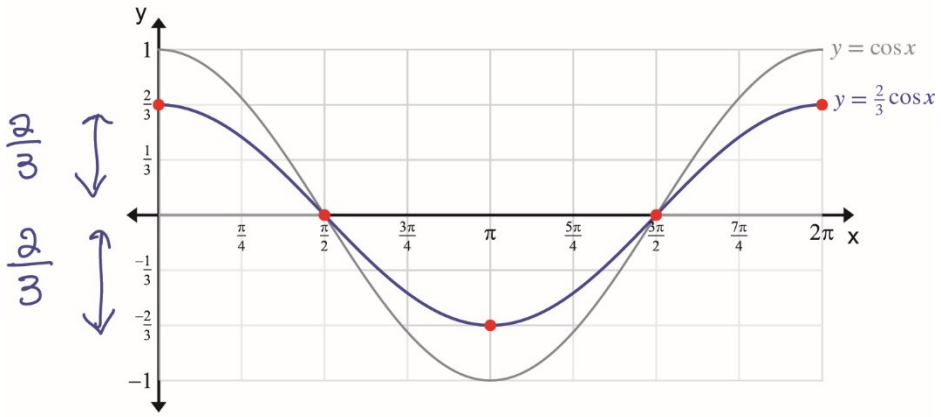
$\text{amp} = \left|\frac{2}{3}\right| = \frac{2}{3}$

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Graph the function.

3. $y = \frac{2}{3}\cos x$

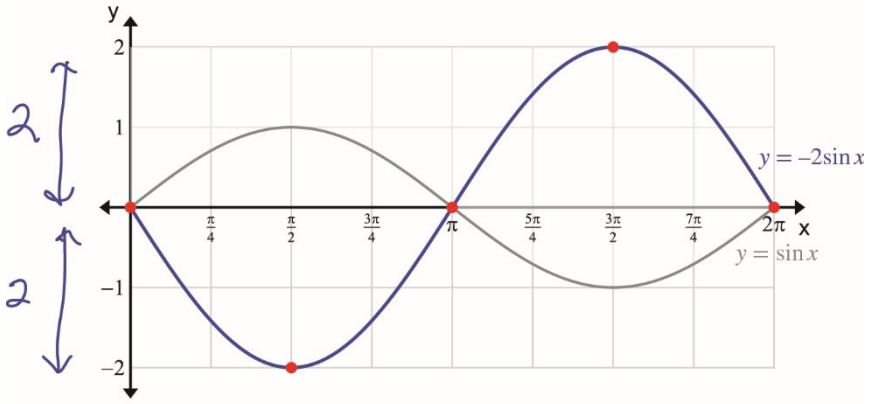
$\text{amp} = \frac{2}{3}$



4. $y = -2\sin x$

reflect across x-axis

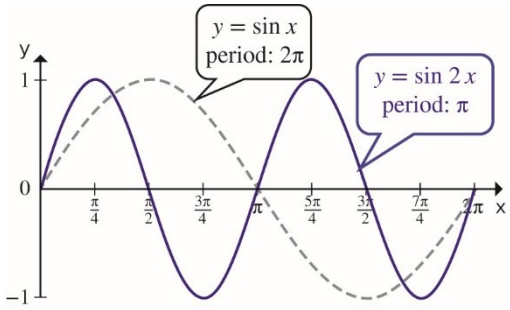
$\text{amp} = |-2| = 2$

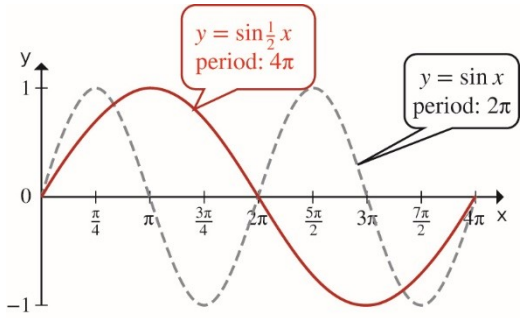


Graph $y = A\sin Bx$ and $y = A\cos Bx$

The graph of $y = f(Bx)$ is the graph of $y = f(x)$ with a horizontal shrink or stretch.

For $y = \sin x$ and $y = \cos x$, the factor B will affect the period of the graph.





To find the interval for one complete cycle of the graphs of $y = \sin Bx$ and $y = \cos Bx$ for $B > 0$, solve the inequality.

For $B > 0$, the period of $y = \sin Bx$ and $y = \cos Bx$ is $\frac{2\pi}{B}$.

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$$0 \le Bx \le 2\pi$$

$$\frac{0}{B} \le \frac{Bx}{B} \le \frac{2\pi}{B}$$

Start value for the cycle

$$0 \le x \le \frac{2\pi}{B}$$

End value for the cycle

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To analyze the period of a sine or cosine function with a negative coefficient on x, for $B > 0$:

Rewrite $y = \sin(-Bx)$ as $y = -\sin Bx$ Rewrite $y = \cos(-Bx)$ as $y = \cos Bx$

Graph the function over one full period and identify the amplitude and period.

5. $y = 2\cos(-4x)$

$y = 2\cos(-4x) = 2\cos 4x$

amp $= A = 2$

$\text{period} = \frac{2\pi}{B} = \frac{2\pi}{4} = \frac{\pi}{2}$

one complete cycle between 0 and $\frac{\pi}{2}$.

divide into fourths $\frac{1}{4}\left(\frac{\pi}{2}\right) = \frac{\pi}{8}$

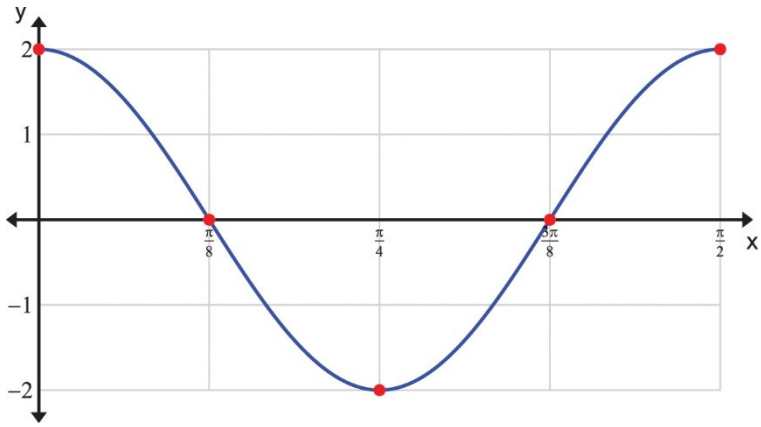
start (0, 2)

(1) $0 + \frac{\pi}{8} = \frac{\pi}{8} \rightarrow \left(\frac{\pi}{8}, 0\right)$

(2) $0 + 2\left(\frac{\pi}{8}\right) = 0 + \frac{\pi}{4} \rightarrow \left(\frac{\pi}{4}, -2\right)$

$0 \le 4x \le 2\pi$

$0 \le x \le \frac{\pi}{2}$



(3) $0 + 3\left(\frac{\pi}{8}\right) = \frac{3\pi}{8} \rightarrow \left(\frac{3\pi}{8}, 0\right)$

(4) $0 + 4\left(\frac{\pi}{8}\right) = \frac{\pi}{2} \rightarrow \left(\frac{\pi}{2}, 2\right)$

Graph $y = A\sin(Bx - C) + D$ and $y = A\cos(Bx - C) + D$

The graph of $y = f(x - h)$ is the graph of $y = f(x)$ with a horizontal shift of $|h|$ units. If $h > 0$ the shift is to the right, and if $h < 0$ the shift is to the left.

$y = A\sin(Bx - C)$ $y = A\cos(Bx - C)$

The graphs of and have both a change in period and a horizontal shift. The horizontal shift to the left or right is called the phase shift.

For $B > 0$, the phase shift can be determined by solving the following inequality.

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$$0 \le Bx - C \le 2\pi$$

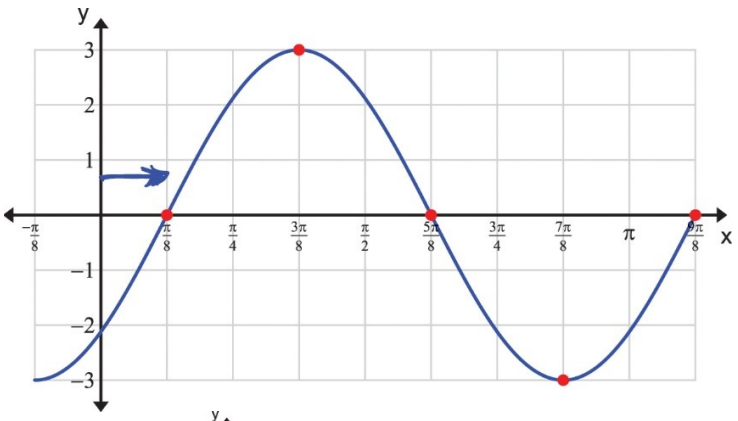
$$C \le Bx \le 2\pi + C$$

The phase shift is the left endpoint.

$$\frac{C}{B} \le x \le \frac{2\pi + C}{B}$$

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6. Graph the function over one full period.
Identify the amplitude, period, and phase shift.



$$y = 3\sin\left(2x - \frac{\pi}{4}\right)$$

$\text{amp} = 3 \qquad \text{period} = \frac{2\pi}{2} = \pi$

one cycle $\quad 0 \le 2x - \frac{\pi}{4} \le 2\pi$

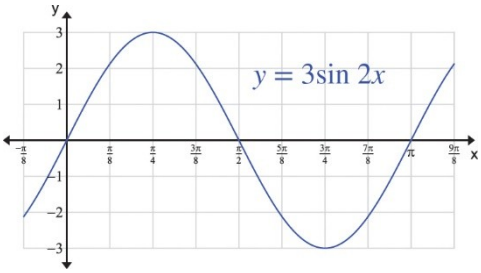
$$\frac{\pi}{4} \le 2x \le 2\pi + \frac{\pi}{4}$$

$$\frac{\pi}{4} \le 2x \le \frac{9\pi}{4}$$

$$\frac{\pi}{8} \le x \le \frac{9\pi}{8}$$

start ↗ ↖ end

phase shift: $\frac{\pi}{8}$ right



$\text{fourths}: \frac{1}{4}(\pi) = \frac{\pi}{4} \quad \text{start} \left(\frac{\pi}{8}, 0\right)$

(1) $\frac{\pi}{8} + \frac{\pi}{4} = \frac{3\pi}{8} \rightarrow \left(\frac{3\pi}{8}, 3\right)$

(2) $\frac{\pi}{8} + 2\left(\frac{\pi}{4}\right) = \frac{5\pi}{8} \rightarrow \left(\frac{5\pi}{8}, 0\right)$

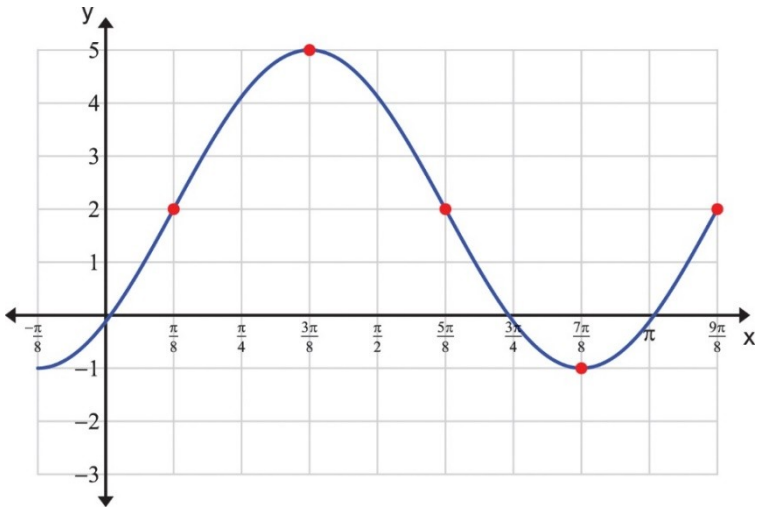
(3) $\frac{\pi}{8} + 3\left(\frac{\pi}{4}\right) = \frac{7\pi}{8} \rightarrow \left(\frac{7\pi}{8}, -3\right)$

(4) $\frac{\pi}{8} + 4\left(\frac{\pi}{4}\right) = \frac{9\pi}{8} \rightarrow \left(\frac{9\pi}{8}, 0\right)$

To find the vertical shift, recall that the graph of $y = f(x) + D$ is the graph of $y = f(x)$ shifted $|D|$ units upwards for $D > 0$ and $|D|$ units downwards if $D < 0$.

7. Graph $y = 3\sin\left(2x - \frac{\pi}{4}\right) + 2$ (range?) $\quad [-1, 5]$

↓ all points up 2 units



$\text{start} \left(\frac{\pi}{8}, 0 + 2\right) = \left(\frac{\pi}{8}, 2\right)$

(1) $\left(\frac{3\pi}{8}, 3 + 2\right) = \left(\frac{3\pi}{8}, 5\right)$ etc.

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Properties of the General Sine and Cosine Functions

For the graphs of $y = A\sin(Bx - C) + D$ and $y = A\cos(Bx - C) + D$ with $B > 0$.

Amplitude = $|A|$ Period = $\frac{2\pi}{B}$ Phase shift = $\frac{C}{B}$ Vertical shift = D

One full cycle is given on the interval $0 \le Bx - C \le 2\pi$. Domain is $-\infty < x < \infty$.

Range is $-|A| + D \le y \le |A| + D$.

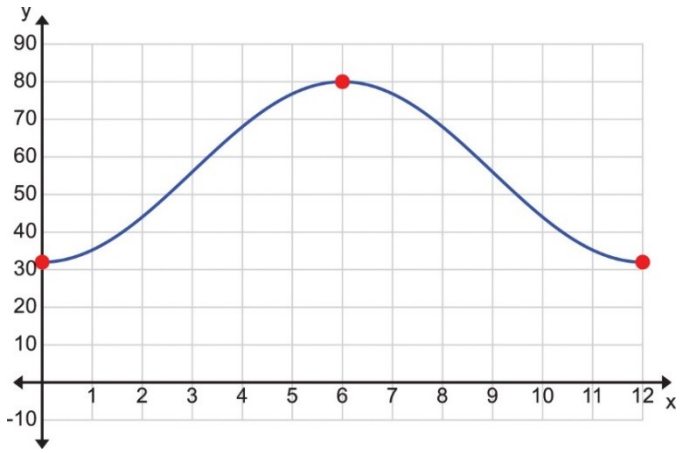
Model Sinusoidal Behavior

8. The monthly high temperature for Buffalo, New York, peaks at an average high of 80° in July down to an average high of 32° in January. Assume that this pattern for monthly high temperatures continues indefinitely and behaves like a cosine wave.

Write a function of the form $H(t) = A\cos(Bt - C) + D$ to model the average high temperature.

$H(t)$

The value is the average high temperature for month t, with January as $t = 0$.



$$\text{Amp} = \frac{\text{high} - \text{low}}{2} = \frac{80 - 32}{2} = 24$$

$$H(t) = -24\cos(Bt - C) + D$$

$$D = \text{vert. shift} = \frac{\text{high} + \text{low}}{2} = \frac{80 + 32}{2} = 56$$

$$H(t) = -24\cos(Bt - C) + 56$$

$$\text{Period} = \frac{2\pi}{B} \rightarrow 12 = \frac{2\pi}{B} \quad B = \frac{2\pi}{12} = \frac{\pi}{6}$$

$$H(t) = -24\cos\left(\frac{\pi}{6}t - C\right) + 56$$

$\rightarrow \quad C = 0 \qquad H(t) = -24\cos\frac{\pi}{6}t + 56$

no phase shift

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